Petek Karagedik

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Education

- **Sabancı University** — İstanbul, Turkey
  BE Computer Science & Engineering + BA Economics (Double Major); GPA: 3.66/4.0 — ***Junior, June 2027***
  - *Dean's Honors List (High Honor: Fall 2023 – Spring 2025)*
  - **Relevant Courses:** Advanced OOP (C++), Data Structures, Algorithms, Database Systems (SQL), Statistics, Machine Learning, Logic & Digital System Design, Linear Algebra, Computational Problem Solving (Python).
- **The University of Hong Kong** — Hong Kong SAR
  Faculty of Engineering Exchange Student (School of Computing and Data Science) — *Fall 2025*
  - **Courses:** Artificial Intelligence, Intro to Data Science, Computer Networks, Interactive Mobile Apps.

Experience

- **Tübitak Bilgem (TUTEL Lab) - Internship** — Jan 2025 – Jan 2025
  Digital Design Intern
  - **Executed advanced digital design projects** using Verilog. Collaborated in a specialized lab environment to design and test circuit protocols, ensuring 100% error-free operation in simulations.
- **Sabancı University - Control Systems Research** — Oct 2023 – Feb 2024
  Student Researcher (Advisor: Prof. Mustafa Ünel)
  - **Investigated modern control architectures** in automotive and defense sectors. Analyzed critical parameters including feedback loops, system stability, and robustness.
  - **Synthesized technical findings** into a comprehensive report and presentation, effectively communicating complex engineering concepts to the research team.
- **DenizBank Overseas Internship - Denizbank** — Mar 2024 – May 2024
  Intern (IT & Management)
  - **Selected from top 10% of applicants** for an intensive program on Agile Transformation and FinTech. Gained cross-functional exposure to Data Science and Blockchain workflows.
- **L'Oréal Türkiye - Campus Brand Advocacy Program** — Oct 2024 – June 2025
  Campus Brand Advocate
  - **Spearheaded campus marketing initiatives** representing the global brand to bridge the gap between beauty and technology sectors. Analyzed student engagement data to optimize messaging.
- **MATH 101 Learning Assistant - Sabancı University** — Feb 2024 – July 2024
  Undergraduate Learning Assistant
  - **Mentored 40 students** in Calculus, demonstrating leadership by breaking down complex technical concepts. Improved class performance through targeted problem-solving sessions.

Projects

- **FPGA Battleship Game Implementation** | *Verilog, Digital Logic Design*
  - **Engineered a digital logic circuit** on an FPGA board to simulate the Battleship game. Designed finite state machines (FSM) for game logic and visual output, ensuring efficient hardware resource usage.
- **HKUnite Event Management Mobile App** | *Java, Android Studio*, SQL, Python
  - **Built a comprehensive mobile application** to manage complex event logistics. Designed an intuitive dashboard for real-time guest tracking, solving user pain points through innovative software solutions.
- **Supply Chain Logistics Optimization Tool** | *C++, Operations Research*
  - **Developed a cost-minimization algorithm** that calculates optimal shipping routes by analyzing variable inputs, simulating logic used in large-scale supply chain operations.
- **AI Hype vs. Market Valuation Analysis** | *R, Tidyverse*
  - **Quantified market sentiment** by developing a data pipeline in R that aggregates search terms into a proprietary "AI Hype Index," merging it with historical stock data to model predictive trends.
- **Personal E-Device Sales Database** | *SQL, Relational Design*
  - **Architected a relational database schema** to streamline inventory tracking. Optimized data retrieval for analytics, enabling efficient querying of device specs and sales performance.
- **High-Frequency Financial Transaction Ledger** | *C++, Blockchain*
  - **Engineered a concurrent transaction simulation** ensuring 100% ledger integrity via SHA-like hashing and mutex synchronization to prevent race conditions.
- **Stochastic Decision Making & Inference Agents** | *Python, Reinforcement Learning*
  - **Modeled Markov Decision Processes (MDPs)** to train autonomous agents using Value Iteration and Q-Learning algorithms. Implemented Epsilon-Greedy strategies to balance exploration and exploitation.
  - **Developed probabilistic tracking systems** using Particle Filters and Bayes Nets to estimate hidden states from noisy sensor data, optimizing inference accuracy in dynamic environments.
- **Autonomous Multi-Agent Search Systems** | *Python, Graph Theory*
  - **Implemented advanced search algorithms** (DFS, BFS, UCS, A*) with custom heuristics to solve complex pathfinding puzzles. Optimized frontier management for memory efficiency.
  - **Designed adversarial game agents** using Minimax and Expectimax algorithms with Alpha-Beta pruning, enabling optimal decision-making against stochastic opponents.
- **Labor Market Automation Research** | *Economic Theory Analysis*
  - **Synthesized economic models** (Acemoglu & Restrepo) to analyze the impact of technology on labor markets, presenting data-driven findings on how automation shifts wage structures.

Extracurricular

- **Economics and Management Club** : Collaborated with a team of 30 and lead teams to coordinate sponsorship with 200+ companies. Secured funding increasing participation by 50% and negotiated partnerships with 50 new firms.
- **TOM @ University Project - Volunteer** : Designed and 3D-printed a toothpaste dispenser for individuals with limited mobility. Published open-source design accessed by hundreds of users globally.

Skills Summary

- **Technical Skills:** C++, Python, SQL, R, Java, Verilog, OOP, Data Structures, Algorithms
- **Tools:** Visual Studio Code, Android Studio, R Studio, MySQL Workbench, FPGA Boards, Github
- **Soft Skills:** Problem Solving, Technical Writing, Leadership, Cross-Functional Collaboration, Adaptability